

Report
Of
Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410030395

Name of Student: ABHISHEK KUMAR SHARMA

Section: 3CSE10

Batch: 2024-28

CANDIDATE'S DECLARATION

I, ABHISHEK KUMAR SHARMA, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Abhishek Kumar Sharma

Roll No.: 2410030395

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

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I express my gratitude to my parents for their blessings.

Signature

Student Name: Abhishek Kumar Sharma

Roll No.: 2410030395

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Abhishek Kumar Sharma

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 47728792361a8513776c

Student ID: STU6997e556103781771562326



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-59 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Data Science has emerged as one of the most sought-after disciplines in the technology industry, enabling organizations to derive actionable insights from vast volumes of data. With India being one of the largest hubs for data science talent globally, hands-on exposure to data engineering, machine learning, and analytical platforms is essential for engineering graduates. This report presents the work carried out during an 8-week Data Science Master Virtual Internship, undertaken through the AICTE–EduSkills National Internship Portal, supported by Siemens (Altair Academy). The internship provided structured, hands-on exposure to data engineering, exploratory data analysis, machine learning, and platform administration using the RapidMiner data science platform, progressing from professional-level to master-level competency across seven modules.

4.2 Organization Profile

EduSkills Foundation is a non-profit organization working under the AICTE (All India Council for Technical Education) umbrella to enable an Industry 4.0-ready digital workforce across India, delivering structured virtual internships to students from AICTE-affiliated and recognized institutions. This particular internship, the Data Science Master Virtual Internship, is supported by Siemens through Altair Academy — Altair Engineering having been acquired by Siemens in 2025 — and is delivered on the RapidMiner platform, a widely used no-code/low-code data science and machine learning tool. The program combines self-paced learning modules, industry-aligned use cases, and assessments to bridge the gap between academic learning and real-world data science practice.

4.3 Problem Statement

Organizations today generate vast amounts of raw, unstructured data, but deriving meaningful, actionable insights from this data requires structured data engineering, rigorous analysis, and reliable predictive models. Many students and early-career professionals, however, lack hands-on exposure to industry-standard data science tools and workflows, relying instead on theoretical classroom knowledge alone. There is a clear need for practical, guided training that

takes a learner through the complete data science lifecycle — from raw data to a validated, deployable model — using a platform and methodology aligned with real industry practice.

4.4 Project Objectives

- To understand the end-to-end data science lifecycle, from data collection to model deployment.
- To gain proficiency in data engineering techniques such as data cleaning, transformation, and preparation.
- To perform exploratory data analysis (EDA) and derive meaningful statistical insights from data.
- To build, validate, and optimize machine learning models using the RapidMiner platform.
- To apply data science techniques to real-world, industry-aligned use cases and applications.
- To gain working knowledge of platform administration — managing workflows, processes, and repositories.
- To progress from professional-level to master-level competency across all seven program modules.

4.5 Scope of the Project

The scope of this internship covers the complete data science workflow taught across the seven modules of the program — spanning applications and use cases, data engineering, and machine learning, each at both professional and master levels, along with platform administration. The learning and practical exercises were carried out using the RapidMiner (Altair AI Studio) platform, covering data preparation, exploratory analysis, model building, and evaluation on representative datasets. The scope does not extend to building a full production-grade enterprise data pipeline or deploying models at large scale, which lies beyond the scope of an 8-week learning-oriented internship.

4.6 Technologies and Tools Used

Category	Tools / Technologies
Data Science Platform	RapidMiner / Altair AI Studio (Altair Academy, Siemens)
Data Engineering	Data cleaning, transformation, and ETL processes
Exploratory Data Analysis	Statistical summaries, data visualization

Category	Tools / Technologies
Machine Learning	Classification, Regression, Clustering algorithms
Model Evaluation	Cross-validation, performance metrics, optimization
Applications & Use Cases	Industry-oriented predictive analytics use cases
Platform Administration	Workflow, process, and repository management in RapidMiner

Table 4.1: Technologies and Tools Used

4.7 System Architecture

The data science workflow followed during the internship begins with raw data collection, followed by data cleaning and preparation to handle missing values, inconsistencies, and formatting issues. This is followed by exploratory data analysis (EDA) to understand patterns, distributions, and relationships within the data, and feature engineering to prepare relevant inputs for modelling. Machine learning models are then built using the RapidMiner/Altair AI Studio platform, followed by rigorous validation and optimization to ensure reliability. The final stage involves deploying the model and translating its output into actionable business insights.

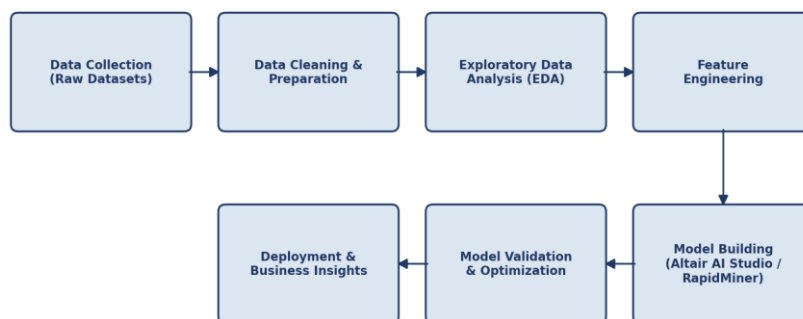


Figure 4.1: Data Science Workflow – From Raw Data to Deployment

4.8 Methodology

The internship followed a structured, module-wise methodology moving from foundational, professional-level concepts to advanced, master-level competency across seven modules, as summarized below.

Module	Focus Area
01	Applications and Use Cases – Professional: introduction to real-world data science applications

Module	Focus Area
02	Data Engineering – Professional: data collection, cleaning, and preparation techniques
03	Machine Learning – Professional: foundational supervised and unsupervised learning algorithms
04	Data Engineering – Master: advanced data transformation and pipeline design
05	Machine Learning – Master: advanced model building, validation, and optimization
06	Applications & Use Cases – Master: industry-specific predictive analytics projects
07	Platform Administrator – Master: managing workflows, processes, and deployments on the RapidMiner platform

Table 4.2: Module-wise Methodology

4.9 Expected Outcomes

- Practical proficiency in data engineering, exploratory data analysis, and machine learning.
- Ability to build, validate, and optimize predictive models using the RapidMiner platform.
- Understanding of real-world, industry-aligned data science applications and use cases.
- Working knowledge of platform administration — managing data science workflows and processes.
- Successful completion of the internship with an ‘Outstanding (O)’ grade certificate awarded by AICTE, EduSkills, and Siemens.

4.10 Certificates of completion and other communication mail proof

The Certificate of Virtual Internship issued jointly by the Ministry of Education, AICTE, the National Internship Portal, and EduSkills (supported by Siemens), confirming successful completion of the 8-week Data Science Master Virtual Internship with an ‘Outstanding (O)’ grade, is attached in Chapter 3 (Internship Completion Certificate) of this report.

CHAPTER 5

BIBLIOGRAPHY/REFERENCES

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